T1 Stress Relaxation in a 2D Bidisperse Emulsion under Wide-Gap Couette Shear: Setup, Protocol, and Verification Document

Working draft / methods section
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We describe the simulation setup for studying T1 (neighbour-swap) plastic events and their associated stress relaxation in a two-dimensional bidisperse emulsion confined to an annular Couette geometry. The system consists of $P = 635$ deformable droplets (50/50 bidisperse, $R_0 \in \{0.8, 1.2\}$) packed at $\phi \approx 0.86$ between concentric circular walls. Shear is injected by a rough inner ring of frozen droplets rotating at angular velocity Ω; an outer pinned ring of frozen droplets prevents rigid co-rotation of the bulk. All quantities required to reconstruct per-pair contact forces post-hoc (positions, velocities, neighbour topology) are saved at 1250-step intervals. This document specifies every numerical value used in the run, identifies the source files implementing each step, and records the validation checks performed during preparation.

I. BACKGROUND AND MOTIVATION

T1 events — topological neighbour swaps in dense disordered packings — are the elementary plastic excitations in two-dimensional foams, emulsions, and granular materials [2, 3]. Under slow shear, a sheared 2D dense system stores elastic energy between rearrangements; each T1 release a finite stress drop localised in space and time. Tracking the spatiotemporal statistics of T1 events and their stress signatures is central to the modern picture of amorphous-solid plasticity [4, 5].

Couette shear is the canonical geometry for this question because (i) the imposed strain rate is well-defined at the moving wall, (ii) the curved geometry localises strain near the inner wall (shear stress decays as $1/r^2$), producing a quasi-1D shear band whose thickness is set by the material rather than the gap [6, 7], and (iii) the boundary conditions are straightforward to implement at the particle level. We follow the established granular-DEM convention of using a *rough* inner wall (a ring of frozen particles co-rotating with the wall) so that strain is injected via particle-particle contacts rather than through a slip-prone smooth boundary [8].

The bidisperse $R_{\text{large}}/R_{\text{small}} = 1.5$ choice follows the Lemaître–Falk tradition for 2D plasticity studies, where size dispersion is essential to suppress hexagonal crystallisation at ϕ near random close packing [4, 5].

II. GEOMETRY AND PARTICLE POPULATION

The accessible domain is the annulus $R_{\text{inner}} < r < R_{\text{outer}}$ centred at (c_x, c_y). All lengths in this paper are reported in units of the mean particle radius $\langle R_0 \rangle = 1$ (the simulator normalises the per-particle R_0 values so $\langle R_0 \rangle = 1$ exactly [*Code:* `src/epd/system.py:initialize`*, the* `assert` *on line 272.*]).

The geometry is parameterised by the design variables

$$N_{\text{inner}} \equiv R_{\text{inner}}/R_0 = \text{inner-radius scale (curvature)} \tag{1}$$

$$N_{\text{width}} \equiv (R_{\text{outer}} - R_{\text{inner}})/(2R_0) = \text{gap, in droplet diameters} \tag{2}$$

together with the target packing fraction ϕ and droplet count P. These four are linked by the constraint

$$\phi \;=\; \frac{P\,\langle r_{\text{eff}}^2 \rangle}{4N_{\text{width}}(N_{\text{inner}} + N_{\text{width}})}, \tag{3}$$

where $r_{\text{eff}} = R_0 + r_{\text{c}}$ is the outer-perimeter radius of one droplet ($r_{\text{c}} = 2\pi R_0/N_{\text{node}}$ is the contact radius) and $\langle\cdot\rangle$ averages over the polydisperse population. Three of $\{N_{\text{inner}}, N_{\text{width}}, \phi, P\}$ determine the fourth [*Design recipe and inversions tabulated in* `docs/couette_design.md`.].

For our run, picking $N_{\text{inner}} \approx 8$ and $N_{\text{width}} \approx 12$ (the "balanced" recipe targeting an inner-wall layer of ≈ 25 driven droplets and a gap wide enough to resolve a shear band of 5–10 diameters [6, 7]) yields $P \approx 635$ at $\phi = 0.84$ via Eq. 3. The realised values after the adaptive compression are reported in Table I.

The Couette cell itself is implemented as two arc walls: a *concave* outer arc at radius R_{outer} (particles inside, pushed inward by the wall potential) and a *convex* inner arc at R_{inner} (particles outside, pushed outward) [*Code:* `src/epd/objects.py:CouetteCell`.]. The arc walls have zero physical radius ($r_{\text{c}}^{\text{wall}} = 0$) but exert linear-spring contact forces on droplet nodes that overlap them.

Particle physics knobs (emulsion mode). Each droplet is a closed elastic membrane with a soft area constraint, following the Elastic Perimeter Disk model [1] [*Full derivation in* `papers/summary_of_methods/main.pdf`.]. Per-droplet physics parameters:

- Line tension $\gamma = 1.0$ (sets the energy unit)
- Area-spring ratio $q = K_{\text{area}}/\gamma R_0 = 5$, fixing $K_{\text{area}} = 5/R_0$
- Contact stiffness $C = 500\,\gamma/R_0$

TABLE I. Geometry and particle parameters of the production setup. Values quoted in units of $\langle R_0 \rangle = 1$. "Initial" quantities are pre-swell user-specified; "post-swell" are the values realised after the adaptive compression (§III).

quantity	symbol	value
particle count	P	635
nodes per droplet	N_{node}	60
size distribution		50% $R_0 = 0.8$ 50% $R_0 = 1.2$
size ratio	$R_{\mathrm{large}}/R_{\mathrm{small}}$	1.5
contact radius (small)	r_c	0.0838
contact radius (large)	r_c	0.1257
inner radius (post-swell)	R_{inner}	7.88
outer radius (post-swell)	R_{outer}	31.53
gap (diameters)	N_{width}	9.5
inner-radius scale	N_{inner}	7.9
radius ratio	$R_{\mathrm{ratio}} = R_{\mathrm{outer}}/R_{\mathrm{inner}}$	4.0
box size (post-swell)	$L_x = L_y$	66.2
target packing fraction	ϕ_{target}	0.84
realised packing fraction	$\phi_{\mathrm{post-swell}}$	0.843

- Ohnesorge number Oh = 0.15 (sets per-node Stokes drag ξ; non-zero drag is required during equilibration to drain kinetic energy)
- Bending modulus negligible ($\tau \to 0$ limit; pure surface tension governs deformation resistance)

For the bidisperse population, the values listed are per-droplet: larger droplets at $R_0 = 1.2$ inherit a smaller K_{area} and C (both proportional to $1/R_0$), so the dimensionless physics (q, CR_0/γ) is identical for both droplet sizes.

III. INITIAL-STATE PREPARATION

The initial state is prepared in three stages.

A. RSA placement

Random sequential addition places $P = 635$ non-overlapping droplet centres in an inflated bounding box of size $L_x^{\mathrm{RSA}} = 105$. The box is sized so that the initial packing fraction in box-area terms is $\phi_{\mathrm{box}}^{\mathrm{RSA}} = 0.25$ [*The simulator auto-expands the box if the user-specified L_x would put $\phi_{\mathrm{box}}^{\mathrm{RSA}}$ above* 0.25. *Code:* `src/epd/system.py:initialize`, *lines 217–229.*]. Droplets whose centres fall outside the user-specified annulus ($R_{\mathrm{inner}}^{\mathrm{initial}} = 12.5$ to $R_{\mathrm{outer}}^{\mathrm{initial}} = 50$) are rejected. Polydisperse R_0 values are sampled from the bimodal distribution before placement, and the simulator enforces $\langle R_0 \rangle = 1.0$ exactly [*Code:* `src/epd/particles.py:ParticleSpec.sample_R0`, *with* `poly_dist={type:'bimodal', ratio:0.5, delta:0.2}`.].

B. Adaptive swell to target ϕ

After RSA, an iterative compression schedule shrinks the box by small fractional steps $\Delta\phi$ (cap $\Delta\phi_{\max} = 0.004$, initial $\Delta\phi_0 = 4 \times 10^{-4}$). Each compression re-scales the box, both arc walls (radii and centre), and all droplet centres by a common factor $\sqrt{\phi_{\mathrm{now}}/\phi_{\mathrm{new}}}$. Following each compression the system relaxes for $n_{\mathrm{relax}} = 400$ time steps under enhanced damping ($\alpha_{\mathrm{swell}} = 15\alpha_{\mathrm{dyn}}$); the run is rolled back and $\Delta\phi$ halved if

1. the maximum capsule overlap fraction f exceeds $f_{\mathrm{crit}} = 0.65$, or
2. the minimum particle circularity $4\pi A/L^2$ falls below 0.30.

The schedule terminates when ϕ_{box} reaches the converted target $\phi_{\mathrm{target}}^{\mathrm{box}} = \phi_{\mathrm{target}} \cdot A_{\mathrm{acc}}/L_x L_y$. At the run-time-imposed $\phi_{\mathrm{target}} = 0.84$, this requires a compression factor ~ 0.63, taking the cell from $(R_{\mathrm{inner}}, R_{\mathrm{outer}}) = (12.5, 50)$ to $(7.88, 31.5)$. The swell took 1395 s wall-time on an RTX 3080 with the integrator at $f_{\mathrm{dt}} = 0.5$ [*Code:* `src/epd/initializer.py:adaptive_swell`.].

C. Equilibrating relax under Stokes drag

After swell completes, the system is integrated for $N_{\mathrm{relax}} = 15{,}000$ steps at $f_{\mathrm{dt}} = 1.5$ with no driving and Stokes drag (Oh = 0.15) active. This drains residual kinetic energy from the swell impulses; in our run the final $\rho_{\mathrm{contact\,relax}}^{\max} = 0$ to fp64 precision, confirming the system is at mechanical rest [*Code:* `src/epd/system.py:run`, *which delegates to* `src/simulation/tf_sim.py:run_simulation_tf`.]. The realised packing fraction drifts slightly upward during relax (here $0.843 \to 0.854$) as the swell-induced internal stresses relax and droplets pack more efficiently.

The post-relax state defines the $t = 0$ reference for the driven phase.

IV. DRIVING PROTOCOL

A. Wall identification

Droplets are assigned to one of three populations based on their post-relax centre-of-mass distance from the cell origin $(c_x, c_y) = (33.1, 33.1)$:

- *Inner ring* (driven, frozen body, count: 25): $|\mathbf{r}| < R_{\mathrm{inner}} + 1.5\langle R_0 \rangle = 9.4$.
- *Outer ring* (pinned, frozen body, count: 85): $|\mathbf{r}| > R_{\mathrm{outer}} - 2\langle R_0 \rangle = 29.5$.

- *Bulk* (free, count: 525): everything in between. The T1 events of interest occur in this population.

The selected ring populations are azimuthally complete (max angular gap 26° inner, 7° outer) so that both rings act as contiguous rough boundaries.

B. Trajectory specification

For each driven droplet i, the simulator advances the centre-of-mass and body angle along a prescribed trajectory $(\mathbf{x}_{\rm cm}^{(i)}(t), \theta^{(i)}(t))$ regardless of internal forces; node-internal coordinates are frozen by the `frozen` flag [*Code:* `src/simulation/tf_sim.py:make_traj`, `src/epd/system.py:set_driven_particles`.].

Inner-ring trajectory. Orbital rotation about the cell centre at angular velocity $\Omega = 2 \times 10^{-3}$ (in units $1/\sqrt{\gamma/\rho R_0^3}$), with body co-rotation matching the orbital rate:

$$\dot{\mathbf{x}}_{\rm cm}^{(i)}(t) = \Omega\, \hat{z} \times (\mathbf{x}_{\rm cm}^{(i)} - \mathbf{c}) \tag{4}$$

$$\dot{\theta}^{(i)}(t) = \Omega \tag{5}$$

The body co-rotation $\dot{\theta} = \Omega$ is necessary so that the rough surface of each driven droplet stays tangent to the annulus as it traverses the inner perimeter; without it the rough side of each driven droplet would slowly turn inward, breaking the "rough wall" boundary condition.

Outer-ring trajectory. Identically zero ($\mathbf{v}_{\rm cm} = 0$, $\dot{\theta} = 0$). These droplets are frozen in place and act as a pinned anchor. Without this anchor, an isolated rough inner ring in a frictionless system would simply spin the bulk rigidly along with itself; the pinned outer ring provides the reaction torque that creates the shear gradient.

C. Time integration and stability watchdog

Integration uses the velocity-Verlet rigid-body decomposition of [1] with global per-step $\Delta t = f_{\rm dt}\, \Delta t_{\rm max}^{\rm CFL}$, where $\Delta t_{\rm max}^{\rm CFL} = 2/\omega_{\rm max}$ is the CFL bound set by the highest-frequency oscillation in the system (here capillary waves on the smallest droplets, $R_0 = 0.8$). We use $f_{\rm dt} = 0.8$ during driving (chosen empirically; $f_{\rm dt} = 1.5$ was found to overshoot to a $\rho_{\rm contact}$ value > 0.4 within 20 chunks at $\phi = 0.86$, indicating a sliding-shear instability).

The closing-rate stability watchdog [1] is monitored every step:

$$\rho_{\rm contact} \;=\; \max_{(a,b)\,\text{in contact}} \frac{\|\Delta\mathbf{x}_a - \Delta\mathbf{x}_b\|}{r_{\rm c}^{(a)} + r_{\rm c}^{(b)}}, \tag{6}$$

the per-contact closing rate normalised by the contact radius. Three thresholds at 0.05 (advisory), 0.10 (strong), 0.20 (critical) provide running visibility into integrator stability. At $f_{\rm dt} = 0.8$ in the production run we observe $\rho_{\rm contact}^{\rm max} \lesssim 0.01$ throughout, more than 5× below the advisory level.

D. Total drive budget

The driven phase spans $N_{\rm drive} = 600{,}000$ steps in 12 segments of 50,000 steps each. At $\Delta t = f_{\rm dt} \cdot 2/\omega_{\rm max} \approx 0.005$ (in units of capillary time), this covers $\Omega T_{\rm drive}/2\pi \approx 1$ rotation of the inner ring.

V. FORCE MODEL AND INTEGRATOR

The full force-model derivation is given in the methods paper [1]. Per-node forces decompose as

$$\mathbf{F}_{i,k}^{\rm node} = \mathbf{F}_{i,k}^{\rm el} + \mathbf{F}_{i,k}^{\rm contact} + \mathbf{F}_{i,k}^{\rm drag} + \mathbf{F}_{i,k}^{\rm reg}, \tag{7}$$

with:

- $\mathbf{F}^{\rm el}$: edge-tension (line tension γ), area-spring (uniform pressure correcting $A - A_0$ via $K_{\rm area}$), and bending forces.

- $\mathbf{F}^{\rm contact}$: pairwise Hertz-type linear penalty, summed over candidate edge–edge pairs from the candidacy manager (see below) plus all primitive walls.

- $\mathbf{F}^{\rm drag}$: per-node Stokes drag $-\xi \mathbf{v}^{\rm node}$, with ξ derived from Oh.

- $\mathbf{F}^{\rm reg}$: a tangential regularisation pseudo-force that prevents node bunching along the perimeter for emulsion droplets only.

The integrator updates centre-of-mass position and rotation as a rigid body, with internal mode displacements $(\mathbf{u}, \dot{\mathbf{u}})$ carried separately [*Code:* `src/simulation/tf_sim.py:step_rb_tf` *(rigid-body update),* `step_full_tf` *(full per-step including contact + watchdog)*.]. The full per-step kernel is JIT-compiled by XLA at the first call; all subsequent steps reuse the compiled cluster.

Candidacy management

A C++ pybind11 extension (`CandidacyManager`) maintains a neighbour matrix `CapCandidates` $\in \mathbb{Z}^{K\times E}$, where $K = P \cdot N_{\rm node} = 38{,}100$ is the per-node row count and $E = 64$ is the candidate-list width. The matrix is rebuilt only when any droplet has moved more than skin/2 since the last rebuild (skin scales with $\langle R_0 \rangle$); typical update cadence in our run is one rebuild per ~ 30 integration steps [*Code:* `src/simulation/candidacy_manager.py`, `src/cpp/candidacy_manager.hpp`.]. Within the integrator loop, the matrix is supplied to the inter-capsule force kernel as a TF tensor.

VI. DATA CAPTURE AND POST-PROCESSING PIPELINE

A. Per-frame capture

Every 1250 steps, a callback dumps the following per-droplet / per-node arrays to memory, accumulating 40 frames per 50,000-step segment:

- Positions: $\mathbf{x}_{\text{all}}(t) \in \mathbb{R}^{P\times N\times 2}$ (per-node world-frame), $\theta(t) \in \mathbb{R}^P$ (body angle).
- Velocities: $\mathbf{v}_{\text{cm}}(t) \in \mathbb{R}^{P\times 2}$ (CM), $\omega(t) \in \mathbb{R}^P$ (body angular velocity), $\dot{\mathbf{u}}(t) \in \mathbb{R}^{P\times N\times 2}$ (internal-mode velocities). These three together reconstruct the per-node velocity $\mathbf{v}^{\text{node}} = \mathbf{v}_{\text{cm}} + \omega \times (\mathbf{x}^{\text{node}} - \mathbf{x}_{\text{cm}}) + R(\theta)\,\dot{\mathbf{u}}$.
- Neighbour topology: $\texttt{CapCandidates}(t) \in \mathbb{Z}^{K\times E}_{\text{uint16}}$, the full candidacy matrix (downcast to `uint16`; max index $K-1 < 2^{16}$).

All arrays are stored at native fp64 (uint16 for the candidacy matrix) to enable bit-exact post-hoc force reconstruction.

B. What is *not* saved per frame

The per-node force breakdown ($\mathbf{F}^{\text{el}}, \mathbf{F}^{\text{contact}}$, etc.) is omitted because each component is a deterministic function of the saved positions, velocities, and per-particle parameters via the same kernel (`step_full_tf`) used during integration. Per-pair contact forces (which are required for the particle-stress tensor) are similarly recomputable from positions and the candidacy matrix.

This decision saves $\sim$ 60% of the disk footprint ($\sim$2.4 GB instead of $\sim$5 GB at 480 frames) and "future-proofs" the dataset: any future analysis that wants a different stress decomposition can use the same raw data without re-running the simulation.

C. Per-segment checkpointing

After each 50,000-step segment, the run atomically writes

- `segment_NNN.npz` (frames + callback data for this segment, $\sim$200 MB),
- `state_drive.npz` (the live integrator state),
- `meta.npz` (segment counter, ring assignments).

A subsequent run of the same script auto-detects the checkpoints, restores the integrator, and resumes from the next segment. This makes the run robust to spot-instance interruption or local process kill.

D. Post-processing path (offline)

Given the saved fields, the planned post-processing pipeline:

1. Compute Delaunay triangulation of $\{\mathbf{x}_{\text{cm}}\}$ at each frame (Python `scipy.spatial.Delaunay`).
2. Identify *T1 candidates* as edge swaps between consecutive Delaunay triangulations. Filter against persistence: a true T1 must persist for some minimum sim-time τ_{T1} ($\sim$ a few capillary times) to exclude oscillatory contact noise.
3. For each T1, recompute the per-pair contact force at each frame in a window $[t-\Delta, t+\Delta]$ around the event, using the same kernel as during the run (positions + candidacy $\rightarrow$ contact forces).
4. Compute the per-particle stress tensor $\sigma^{(i)}_{\alpha\beta} = (1/A_i)\sum_{j\in\text{neighbours}} r_{ij,\alpha}\, F_{ij,\beta}$, and its $(\operatorname{tr}\sigma)/2$ trace (pressure) and shear component.
5. Track the stress relaxation trajectory in the local region around each T1 to characterise the avalanche size and decay timescale.

The post-processing is implemented as a separate Jupyter notebook that reads only `drive_data.npz` (or the per-segment files) and writes derived quantities; the simulation itself never re-runs.

VII. VALIDATION CHECKS PERFORMED DURING PREPARATION

Geometry recipe consistency. After swell, $R_{\text{inner}} = 7.88$, $R_{\text{outer}} = 31.53$, giving $N_{\text{inner}} = 7.88$ and $N_{\text{width}} = 9.43$. Substituting into Eq. 3 with the per-population $\langle r^2_{\text{eff}}\rangle = 1.27$, the predicted P is

$$P_{\text{pred}} = 4 \cdot 0.843 \cdot 9.43 \cdot 17.31/1.27 \approx 433$$

versus the realised $P = 635$. The $\sim$ 47% discrepancy is absorbed by particle deformation: at $\phi = 0.84$ droplets are slightly squished from the rest area πr^2_{eff}, so the *simulated* packing fraction (Eq. 3 rhs evaluated on actual node-perimeter polygons) over-estimates the relevant "reff". Cross-checked: the simulator's directly computed ϕ_{sim} from outer-perimeter polygon areas matches the adaptive-swell terminus to within 0.005.

Auto-expand sizing. RSA was attempted in the user-specified $L_x = 105$ box; since $P \cdot \pi r^2_{\text{eff}}/L_x^2 = 0.230 < 0.25$, no auto-expand occurred and RSA placed in the original box. This was not a hidden geometry change; the cell remained at $R_{\text{inner}} = 12.5, R_{\text{outer}} = 50$ until the swell phase began compression.

Wall-ring populations. At post-relax, the inner-ring population is 25 droplets, with max angular gap $= 26°$. This is consistent with the geometric estimate $\pi R_{\text{inner}}/R_0 \approx 24.7$ to within the counting noise of the threshold criterion. The outer ring is 85 droplets with max gap $= 7°$, far denser than the inner ring (consistent with the larger circumference $2\pi R_{\text{outer}}$).

Equilibration before driving. After the 15,000-step relax phase, $\rho^{\max}_{\text{contact}} = 0$ to fp64 precision, indicating the system has fully drained to mechanical rest under the imposed Stokes drag. Initial driving impulses therefore act on an unambiguous reference state.

Post-driving stability. At $f_{\text{dt}} = 0.8$, $\rho^{\max}_{\text{contact}} \lesssim 0.01$ throughout the driven phase, well below the 0.05 advisory threshold. No watchdog crossings occurred.

Restart correctness. After an unrelated bug forced an aborted run during phase 3, re-launching the script auto-detected `state_post_swell.npz`, re-initialised the system to the stored swell terminus, and re-ran phase 2 (relax). The realised post-relax state matched bit-for-bit (within fp64 ordering) the original run's, confirming the resume path is deterministic.

VIII. RUN-TIME SUMMARY

The full pipeline (RSA → swell → relax → drive → final render) totals ~ 5 hours wall-clock on a consumer RTX 3080 in fp64. Approximately 80% of that time is the driven phase (600,000 steps, ~ 30 ms/step at $K = P \cdot N_{\text{node}} = 38{,}100$ contact rows). Per-step cost is fp64-saturated on this GPU; a data-centre A100 with native fp64 hardware is expected to reduce per-step cost by $\sim 5\times$.

IX. DISCUSSION: DESIGN CHOICES AND LIMITATIONS

The choice of $R_{\text{ratio}} = 4$ (gap = 9.5 diameters) is on the lower end of the wide-gap range used in granular Couette literature [6, 7]; this was a compromise between bulk depth and per-step compute cost. A wider gap would localise the shear band more cleanly, at the cost of $P \propto (R^2_{\text{ratio}} - 1)$ more droplets. The current configuration resolves a ~ 5-diameter shear band with an additional ~ 4 diameters of "bulk" before the pinned outer boundary.

The $\Omega = 2 \times 10^{-3}$ driving rate is in the quasi-static regime: the inelastic time $\tau_{\text{inel}} = \alpha^{-1}_{\text{damp}} \approx 1$ (in capillary units) is much shorter than the strain time $1/\Omega = 500$, so the system relaxes between successive T1 events. Increasing Ω would push toward the inertial regime and dilute the T1 signature with continuous shear flow; we plan to revisit this scaling in a separate sweep.

The pinned outer ring is necessary because the simulator's walls are frictionless — the outer arc exerts only a normal contact force on overlapping droplets and cannot tangentially constrain them. Without the outer pinned droplets, the bulk would simply co-rotate rigidly with the inner driven ring, and no shear gradient would develop. This is a physical (not numerical) feature: real Couette rheometers with smooth outer walls likewise show co-rotation [9].

X. CODE AVAILABILITY

The simulation code is available at KD-physics/Squishy-Particle-Simulator, v2 branch. The exact run script for this study is provided as `notebooks/getting_started.ipynb` (Test C section, modified for the production parameters of Table I). The geometry recipe is documented in `docs/couette_design.md`.

[1] Methods paper: *The Elastic Perimeter Disk Model: Construction, Parameterization, and Calibration.* Companion document, `papers/summary_of_methods/main.pdf`.

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